Higher-Cell Product Retractions, Simplicial Cochains, and Finite Retraction Dynamics

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Abstract

Product-of-marginals retraction gives a finite diagnostic for departure from a fully factorized quantum-state model. At a product reference state, its derivative is an idempotent oblique projector that preserves every single-site marginal. We distinguish its complementary residual from a different projector onto the highest-order tensor component. Single-site Hamiltonian terms are tangent, whereas a two-site interaction can generate a nonzero product-state residual inside a three-site cell. A worked counterexample corrects the stronger isolation claim in the earlier candidate.

For a finite, closed system with product initial data and a bound on the operator norm of the entire Hamiltonian, we derive a quadratic lower bound on distance to the product of marginals. Lemma 4.3a refines that coefficient by the site budgets of a declared interaction decomposition. Lemma 4.3b carries the same coefficient to the reduced state on a declared subsystem of a finite dilation. The global norm remains in the coefficient. Lemma 4.3c treats one jump generator with no finite autonomous Hermitian dilation. Lemma 4.3d gives a coarser quadratic coefficient for a finite family of jumps and no Hamiltonian; on the Lemma 4.3c data the coefficient is 10, not 6. Lemma 4.3e puts a bounded Hamiltonian beside that family. The quadratic coefficient is $2(1 + k^2)(g + G)^2$. On a commuting witness with $g = G = 1$ and $k = 2$ it equals 40, not 6. Lemma 4.3f shows that the state-acceleration bound used for that square cannot replace $(g + G)^2$ by $g^2 + G^2$: a product witness has $\|\rho''(0)\|_1 = 10 > 8$, while $D(t) = 0$. This distance measures total failure of factorization, including classical correlations; nonzero distance alone does not establish entanglement or genuine multipartite entanglement. For two qubits, however, distance greater than one is sufficient for entanglement (Hall, 2013). We then give the explicit observation needed to turn residuals into simplicial cochains, review finite scalar Hodge decomposition, fix a graded-square sign convention, and derive a separate state-local semigroup defect coefficient. No identity between these two defects is assumed. The package supplies numerical examples and conditional abstract Lean proofs of projector identities. Evidence scope and unfinished formal replay obligations are stated explicitly.

1. Scope, notation, and contribution

All Hilbert spaces, matrices, simplicial complexes, and examples in this paper are finite. The contribution is a consistent account of which projection diagnostic answers which question, with proofs, counterexamples, and a reproducible computational package. The basic projector and Hodge identities are standard finite algebra; this paper does not claim a new complete theory of topology or non-Abelian geometry.

Two indices must remain separate:

Symbol	Meaning	Three-site example
k	Number of quantum subsystems in a support σ	$k = 3$
p	Simplicial dimension and cochain degree	$p = k - 1 = 2$
q	An independently declared memory grade	Not a subsystem count
Q	Complement of the fully factorized tangent projector	Retains two-site and three-site components
R_k	Highest-order reference-dependent component projector	Removes proper-support components

An abstract p-simplex has $p + 1$ vertices. A hyperedge of cardinality k does not by itself supply a simplicial boundary: one must declare a complex containing its faces.

Here a **finite autonomous Hermitian dilation** means one fixed finite-dimensional total carrier, one fixed time-independent Hermitian Hamiltonian, and one fixed initial total state whose reduction reproduces the entire specified trajectory for every $t \geq 0$. The no-go statements in Lemmas 4.3c and 4.3f use this meaning. They do not address time-by-time dilations or unrestricted time-dependent Hamiltonians.

The paper assigns no operational authorization or institutional standing. A mathematical witness, an integrity check, and a policy decision have different meanings.

2. Product retraction and its derivative

Let $\sigma = \{1, \dots, k\}$, with $k \geq 2$, and let

$$\mathcal{H}_\sigma = \bigotimes_{j=1}^{k} \mathcal{H}_j, \qquad \dim \mathcal{H}_j = d_j < \infty.$$

Write $\tau_j Z = \mathrm{Tr}_{\sigma \setminus \{j\}} Z$. Density matrices are positive semidefinite Hermitian matrices of trace one. The ambient affine space is the trace-one Hermitian space; its translation space $\mathcal{B}_0$ consists of traceless Hermitian matrices.

For a density matrix S, define

$$\Pi(S) = \bigotimes_{j=1}^{k} \tau_j S.$$

This map preserves the single-site marginals and fixes every product density. At a reference $\rho = \bigotimes_j \rho_j$, with $\operatorname{Tr} \rho_j = 1$, write

$$\iota_j z_j = \rho_1 \otimes \cdots \otimes \rho_{j-1} \otimes z_j \otimes \rho_{j+1} \otimes \cdots \otimes \rho_k.$$

For $Z \in \mathcal{B}_0$, differentiation on the trace-one affine space gives

$$PZ = D\Pi_\rho Z = \sum_{j=1}^{k} \iota_j(\tau_j Z), \qquad Q = I - P.$$

2.1 Boundary states and the full-space extension

For full-rank factors, the product densities form a smooth manifold locally. At pure or rank-deficient factors, positivity introduces a boundary. At any product reference, $\operatorname{im} P$ is the tangent space of the smooth algebraic product manifold formed from normalized Hermitian factors; $\mathcal{B}_0$ is the ambient variation space. It is not the set of all feasible two-sided density-matrix velocities at a boundary point.

The software accepts arbitrary matrix perturbations through a separately declared homogeneous extension, defined near $\operatorname{Tr} S = 1$:

$$\widehat{\Pi}(S) = (\operatorname{Tr} S)^{1-k} \bigotimes_j \tau_j S.$$

Its derivative is

$$\widehat{P}Z = \sum_j \iota_j(\tau_j Z) - (k-1)\operatorname{Tr}(Z)\rho.$$

It restricts to P on trace-zero directions. Its full-space image contains the radial direction ρ as well as the algebraic tangent space. The correction term is not the unrestricted derivative of the unnormalized map $S \mapsto \bigotimes_j \tau_j S$. The code applies it linearly, without a trace-dependent numerical threshold.

3. Projector identities and the locality correction

Theorem 4.1 (Idempotence, marginals, and kernel)

On $\mathcal{B}_0$,

$$P^2 = P, \qquad \tau_j(PZ) = \tau_j Z,$$

and

$$\operatorname{im} P = \left\{ \sum_j \iota_j z_j : \ z_j = z_j^\dagger, \ \operatorname{Tr} z_j = 0 \right\}, \qquad \ker P = \{Z : \tau_j Z = 0 \text{ for every } j\}.$$

The image is the algebraic product tangent space described in Section 2.1.

Proof. For trace-zero z_j, normalization of the other factors gives

$$\tau_l(\iota_j z_j) = \begin{cases} z_j, & l = j, \\ 0, & l \neq j. \end{cases}$$

Since $\operatorname{Tr} \tau_j Z = \operatorname{Tr} Z = 0$, this identity immediately gives marginal preservation. Applying it twice gives $P^2 Z = PZ$. Each PZ lies in the displayed sum, and P fixes every vector in that sum. Finally, $PZ = 0$ implies $\tau_j Z = \tau_j(PZ) = 0$; the converse follows from the definition. □

Consequently, $Q^2 = Q$, $PQ = QP = 0$, and every single-site marginal of QZ vanishes. The term "normal" refers to the chosen complementary subspace; an oblique P need not define an orthogonal projection.

Theorem 4.2 (One-site tangent isolation)

For a Hermitian operator h_j acting only on site j,

$$Q(-i[h_j \otimes I_{\sigma\setminus\{j\}}, \rho]) = 0.$$

Proof. The commutator is $\iota_j(-i[h_j, \rho_j])$, whose local factor has trace zero. Theorem 4.1 fixes this elementary tangent vector. Linearity also annihilates any sum of one-site commutators. □

This theorem does not extend to every interaction with support smaller than k. For $X \subsetneq \sigma$,

$$-i[h_X \otimes I, \rho] = -i[h_X, \rho_X] \otimes \rho_{\sigma\setminus X},$$

but $-i[h_X, \rho_X]$ need not be a sum of one-site variations.

Counterexample 4.2a (Two-site residual inside a triangle)

Let $k = 3$, $\rho = |000\rangle\langle 000|$, and $H = X_1X_2 \otimes I_3$, where X is Pauli X. Then

$$L = -i[H, \rho] = -i(|110\rangle\langle 000| - |000\rangle\langle 110|).$$

Every single-site marginal of L is zero. Thus

$$PL = 0, \qquad QL = L, \qquad \|QL\|_1 = 2.$$

The support has size two, strictly smaller than three. The spectator-site marginal is zero, but that fact does not establish $QL = 0$. At later times the state is a correlated two-qubit state tensored with $|0\rangle\langle 0|$; it has no genuine three-party entanglement.

Full support is also insufficient for a nonzero residual: any Hamiltonian commuting with the reference state has $L = 0$.

Proposition 4.2b (A distinct highest-order projector)

Let E_jZ replace subsystem j by ρ_j, retaining the reduced operator on all other subsystems, with tensor factors restored to their declared order:

$$E_jZ = \rho_j \otimes_j \operatorname{Tr}_j Z, \qquad R_k = \prod_{j=1}^{k}(I - E_j).$$

Here $\otimes_j$ means insertion at site j. The E_j are commuting idempotents: $E_j^2 = E_j$ and $E_iE_j = E_jE_i$. Therefore $R_k^2 = R_k$. If $Z = Z_X \otimes \rho_{\sigma\setminus X}$ for a proper subset X, choose $j \notin X$. Then $E_jZ = Z$, so $R_kZ = 0$. This establishes proper-support annihilation for R_k.

The highest-order projector selects the tensor component with a traceless local factor at every site. It is reference-dependent and is not a connected information measure; its nonzero value alone is not an entanglement witness. For $k = 2$, it agrees with Q on trace-zero perturbations. For $k > 2$, Q also retains intermediate orders. In Counterexample 4.2a, $R_3L = 0$ while $QL = L$.

4. Short-time distance and dual witnesses

Theorem 4.3 (A finite-Hamiltonian lower bound)

Let $H = H^\dagger$ act on the closed finite system $\mathcal{H}_\sigma$. Assume $\|H\|_\infty \le g$, $g > 0$, and product initial data $\rho(0) = \rho$. Set

$$\rho(t) = e^{-itH}\rho e^{itH}, \quad D(t) = \|\rho(t) - \Pi(\rho(t))\|_1, \quad \nu = \|Q(-i[H,\rho])\|_1.$$

For every $t \ge 0$,

$$D(t) \ge \nu t - 2(1+k^2)g^2t^2 \ge \nu t - 6k^2g^2t^2.$$

If $\nu > 0$ and $0 \le t \le t_* = \nu/(12k^2g^2)$, then $D(t) \ge \nu t/2$, strictly positive for $t > 0$.

Proof. Unitary invariance of the trace norm and the commutator estimate $\|[H,Z]\|_1 \le 2\|H\|_\infty\|Z\|_1$ imply uniformly in time

$$\|\rho'(t)\|_1 \le 2g, \qquad \|\rho''(t)\|_1 \le 4g^2.$$

Partial trace is trace-norm contractive on Hermitian inputs, so each single-site marginal obeys the same two bounds. Every marginal density has trace norm one. Differentiating $F(t) = \bigotimes_j \rho_j(t)$, and using trace-norm multiplicativity on tensor products, yields

$$\begin{aligned}\|F''(t)\|_1 &\le \sum_j \|\rho_j''(t)\|_1 + \sum_{i\ne j}\|\rho_i'(t)\|_1\|\rho_j'(t)\|_1 \\ &\le 4kg^2 + 4k(k-1)g^2 = 4k^2g^2.\end{aligned}$$

For $f(t) = \rho(t) - F(t)$, we have $f(0) = 0$, $f'(0) = n = QL$, and $\|f''(t)\|_1 \le 4(1+k^2)g^2$. The integral remainder gives

$$f(t) = tn + \int_0^t (t-s)f''(s)\,ds, \qquad \|f(t) - tn\|_1 \le 2(1+k^2)g^2t^2.$$

The reverse triangle inequality establishes the first bound; the second is weaker for $k \ge 2$. The stated window makes the weaker quadratic term at most $\nu t/2$. □

Here g bounds the **whole Hamiltonian**, not $\max_j \sum_{X\ni j}\|h_X\|_\infty$. An interaction decomposition supplies a valid, possibly loose choice $g = \sum_X \|h_X\|_\infty$. Lemma 4.3a is the closed-system refinement that uses the site budgets of a declared decomposition. The global norm remains in the coefficient. Lemma 4.3b extends that coefficient to the reduced state of a declared subsystem inside a finite dilation. Lemma 4.3c is one jump trajectory with no finite autonomous Hermitian dilation. Lemma 4.3d gives a quadratic coefficient for a finite family of jumps and no Hamiltonian. On the Lemma 4.3c data that coefficient is 10, not the scalar coefficient 6. Lemma 4.3e puts a bounded Hamiltonian beside that family. The quadratic coefficient is $2(1+k^2)(g+G)^2$. It is not the scalar coefficient 6. Lemma 4.3f shows that the acceleration estimate producing the square cannot replace $(g+G)^2$ by g^2+G^2. The defect coefficient is unchanged.

Lemma 4.3a (Site-budget refinement on a closed system)

Let

$$H = \sum_{\alpha=1}^N h_\alpha$$

on the closed finite system $\mathcal{H}_\sigma$, where each $h_\alpha = h_\alpha^\dagger$ is supported on a declared subset $X_\alpha \subseteq \sigma$ and is extended by the identity on the complement. Let $g \geq \|H\|_\infty$ and

$$g_j = \sum_{\alpha:\, j \in X_\alpha} \|h_\alpha\|_\infty.$$

Keep the product initial state, D, ν, and $f(t) = \rho(t) - \bigotimes_j \rho_j(t)$ of Theorem 4.3. Set

$$\Gamma = g^2 + g \sum_{j=1}^{k} g_j + \sum_{i \neq j} g_i g_j.$$

Then, for every $t \geq 0$,

$$D(t) \geq \nu t - 2\Gamma t^2.$$

If $\nu > 0$ and $0 \leq t \leq \nu/(4\Gamma)$, then $D(t) \geq \nu t/2$.

On the one-term decomposition $h_1 = H$, with declared support $X_1 = \sigma$ and $g = \|H\|_\infty$, one has $g_j = g$ and $\Gamma = (1 + k^2)g^2$, so the quadratic coefficient equals the leading coefficient of Theorem 4.3. A decomposition with some $g_j < g$ can make Γ strictly smaller. A redundant decomposition can make Γ larger. Theorem 4.3 remains available unchanged.

Proof. If $K = I_j \otimes A$ on $\mathcal{H}_j \otimes \mathcal{H}_R$, with $R = \sigma \setminus \{j\}$, then $\tau_j([K, Z]) = 0$ for every matrix Z. In a product basis the kept matrix element is

$$\sum_b [K, Z]_{(ab),(a'b)} = \sum_{b,b''} A_{bb''} Z_{(ab''),(a'b)} - \sum_{b,b''} Z_{(ab),(a'b'')} A_{b''b}.$$

Renaming $b'' \leftrightarrow b$ in the second sum produces the first, because the scalar matrix elements of A commute with those of Z. Terms of H supported off site j therefore drop out:

$$\tau_j([H, Z]) = \tau_j\Big(\Big[\sum_{j \in X_\alpha} h_\alpha,\, Z\Big]\Big).$$

The commutator estimate $\|[B, Z]\|_1 \leq 2\|B\|_\infty \|Z\|_1$ and trace-norm contractivity of the partial trace on Hermitian inputs give $\|\rho'(t)\|_1 \leq 2g$ and $\|\rho''(t)\|_1 \leq 4g^2$, as in Theorem 4.3. The same two facts, applied only to the terms that meet site j and using $\|\rho(t)\|_1 = 1$, give $\|\rho_j'(t)\|_1 \leq 2g_j$. Since $\rho_j'' = \tau_j(-i[H, \rho'])$ and $\|\rho'(t)\|_1 \leq 2g$,

$$\|\rho_j''(t)\|_1 \leq \sum_{j \in X_\alpha} 2\|h_\alpha\|_\infty \|\rho'(t)\|_1 \leq 4g g_j.$$

Every marginal density has trace norm one. The product rule and trace-norm multiplicativity give

$$\begin{aligned} \|F''(t)\|_1 &\leq \sum_j 4g g_j + \sum_{i \neq j} (2g_i)(2g_j) \\ &= 4g \sum_j g_j + 4 \sum_{i \neq j} g_i g_j. \end{aligned}$$

Hence $\|f''(t)\|_1 \leq 4\Gamma$. As in Theorem 4.3, $f(t) = tn + \int_0^t (t - s) f''(s)\, ds$, so the remainder has trace norm at most $4\Gamma \cdot t^2/2 = 2\Gamma t^2$. The reverse triangle inequality gives the bound. If $\nu > 0$, then $\|H\|_\infty > 0$, so $g > 0$ and $\Gamma \geq g^2 > 0$. The stated window makes the quadratic term at most $\nu t/2$. The one-term identities are substitutions. □

The factor g in Γ is the global operator-norm budget. The lemma does not replace it by $\max_j g_j$. If the state on σ is only the reduction of a larger closed system, this closed-system argument does not apply by itself. Lemma 4.3b treats that reduction when a finite autonomous Hermitian dilation is declared. Neither lemma removes g.

Example 4.3a (A strictly sharper coefficient)

Take $k = 3$, $\rho = |000\rangle\langle 000|$, and $H = X \otimes X \otimes I$, so $\|H\|_\infty = 1$. Use the one-term decomposition supported on $\{1, 2\}$, with $g = 1$, $g_1 = g_2 = 1$, and $g_3 = 0$. Then

$$\sum_j g_j = 2, \qquad \sum_{i \neq j} g_i g_j = 2, \qquad \Gamma = 5.$$

The quadratic coefficient is $2\Gamma = 10$. Theorem 4.3 with the same g has leading coefficient $2(1 + k^2)g^2 = 20$. The exact state is

$$|\psi(t)\rangle = \cos t\, |000\rangle - i \sin t\, |110\rangle.$$

The site-3 marginal is $|0\rangle\langle 0|$ for every t, so its velocity has trace norm $0 = 2g_3$. The site-1 marginal is $\mathrm{diag}(\cos^2 t, \sin^2 t)$, with

$$\|\rho_1'(t)\|_1 = 2|\sin(2t)|,$$

equal to $2g_1$ at $t = \pi/4$. Counterexample 4.2a gives $\nu = 2$. Thus $D(t) \geq 2t - 10t^2$, and $D(t) \geq t$ for $0 \leq t \leq 1/10$. The published Theorem 4.3 window on these data, using the weaker coefficient $54g^2$, is only $0 \leq t \leq 1/54$.

This example is an exact symbolic witness. It is not a Python test, and it does not evaluate $D(t)$ itself. The code path for Theorem 4.3 is unchanged and still uses the weaker coefficient.

Lemma 4.3b (Reduced site-budget bound on a finite dilation)

Let the closed finite carrier be the disjoint union $\sigma \sqcup \varepsilon$, with $k = |\sigma| \geq 2$ and ε nonempty. Let

$$H = \sum_{\alpha=1}^{N} h_\alpha$$

be Hermitian on $\mathcal{H}_\sigma \otimes \mathcal{H}_\varepsilon$, where each $h_\alpha = h_\alpha^\dagger$ is supported on a declared subset X_α of the full site set and is extended by the identity off that subset. Let $g \geq \|H\|_\infty$ and, for each system site $j \in \sigma$,

$$g_j = \sum_{\alpha:\, j \in X_\alpha} \|h_\alpha\|_\infty.$$

A term supported only on ε contributes to no system budget g_j. Let

$$\rho_{\mathrm{tot}}(t) = e^{-itH} \rho_{\mathrm{tot}}(0) e^{itH}, \qquad \rho_\sigma(t) = \tau_\varepsilon(\rho_{\mathrm{tot}}(t)),$$

and write $\rho_j(t)$ for the single-site marginal of $\rho_\sigma(t)$. Assume $\rho_\sigma(0) = \bigotimes_{j \in \sigma} \rho_j(0)$. Set $f(t) = \rho_\sigma(t) - \bigotimes_{j \in \sigma} \rho_j(t)$,

$$D_\sigma(t) = \|f(t)\|_1, \qquad \nu_\sigma = \|f'(0)\|_1,$$

and

$$\Gamma = g^2 + g \sum_{j \in \sigma} g_j + \sum_{\substack{i,j \in \sigma \\ i \neq j}} g_i g_j.$$

Then, for every $t \geq 0$,

$$D_\sigma(t) \geq \nu_\sigma t - 2\Gamma t^2.$$

If $\nu_\sigma > 0$ and $0 \leq t \leq \nu_\sigma/(4\Gamma)$, then $D_\sigma(t) \geq \nu_\sigma t/2$.

If instead $H = H_\sigma \otimes I_\varepsilon + I_\sigma \otimes H_\varepsilon$, then for every initial total state

$$\rho_\sigma(t) = e^{-itH_\sigma}\rho_\sigma(0)e^{itH_\sigma}.$$

The norm of H_ε does not appear in this trajectory. When $\rho_\sigma(0)$ is a site product and $g \geq \|H_\sigma\|_\infty$, Lemma 4.3a applies on σ with the site budgets of H_σ alone.

The factor g remains in Γ. The lemma does not replace it by $\max_j g_j$. A reduced generator supplied only as a Lindblad map, without a finite autonomous Hermitian dilation and a declared norm bound, is not covered. An environment-only summand can make the admitted g larger than the norm of a particular reduced generator.

Proof. Lemma 4.3a's single-site identity, applied on the dilated carrier, gives $\tau_j([H, Z]) = \tau_j([\sum_{j\in X_\alpha} h_\alpha, Z])$ for every matrix Z. Partial trace is trace-norm contractive on Hermitian inputs, and $\|[B, Z]\|_1 \leq 2\|B\|_\infty\|Z\|_1$. Hence $\|\rho'_{\mathrm{tot}}(t)\|_1 \leq 2g$ and $\|\rho''_{\mathrm{tot}}(t)\|_1 \leq 4g^2$, and the contracted system derivatives obey the same two bounds. Keeping only the terms that meet system site j, and using $\|\rho'_{\mathrm{tot}}(t)\|_1 \leq 2g$, gives $\|\rho'_j(t)\|_1 \leq 2g_j$ and $\|\rho''_j(t)\|_1 \leq 4gg_j$.

Every system marginal is a density matrix. Differentiating $F(t) = \bigotimes_{j\in\sigma} \rho_j(t)$ and using trace-norm multiplicativity, as in Theorem 4.3, yields

$$\|F''(t)\|_1 \leq \sum_{j\in\sigma} 4gg_j + \sum_{\substack{i,j\in\sigma \\ i\neq j}} 4g_ig_j.$$

Thus $\|f''(t)\|_1 \leq 4\Gamma$. The integral remainder gives $f(t) = tf'(0) + \int_0^t (t-s)f''(s)\,ds$, so the remainder has trace norm at most $2\Gamma t^2$. The reverse triangle inequality is the bound. If $\nu_\sigma > 0$, then $\|H\|_\infty > 0$, so $g > 0$ and $\Gamma \geq g^2 > 0$. The stated window makes the quadratic term at most $\nu_\sigma t/2$.

For the decoupled identity, $H_\sigma \otimes I$ commutes with $I \otimes H_\varepsilon$, so $e^{-itH} = e^{-itH_\sigma} \otimes e^{-itH_\varepsilon}$. In a product basis, conjugation by $I \otimes V$ does not change the partial trace over ε. The summed matrix element is

$$\sum_b \sum_{c,c'} V_{bc} Z_{(ac),(a'c')} V^*_{bc'} = \sum_c Z_{(ac),(a'c)},$$

because $\sum_b V^*_{bc'} V_{bc} = \delta_{c'c}$. Therefore $\tau_\varepsilon((U \otimes V)Z(U^\dagger \otimes V^\dagger)) = U\,\tau_\varepsilon(Z)\,U^\dagger$. Applied to $\rho_{\mathrm{tot}}(0)$, this is the reduced unitary orbit. Lemma 4.3a applies on that orbit, and $\|H_\varepsilon\|_\infty$ is not an input. □

Example 4.3b (A non-unitary reduced witness)

Take $\sigma = \{1, 2, 3\}$, $\varepsilon = \{4\}$, and

$$H = (X \otimes X \otimes I) \otimes |0\rangle\langle 0|,$$

so $\|H\|_\infty = 1$. Use the one-term decomposition supported on $\{1, 2, 4\}$, with $g = 1$, $g_1 = g_2 = 1$, and $g_3 = 0$. Then

$$\sum_{j\in\sigma} g_j = 2, \qquad \sum_{\substack{i,j\in\sigma \\ i\neq j}} g_ig_j = 2, \qquad \Gamma = 5.$$

The quadratic coefficient is $2\Gamma = 10$. Theorem 4.3 on a closed three-site system with the same g has leading coefficient $2(1 + k^2)g^2 = 20$.

Let $|\Psi(0)\rangle = (|0000\rangle + |0001\rangle)/\sqrt{2}$. The system reduction is $|000\rangle\langle 000|$. Here $H|0001\rangle = 0$, while on the environment-$|0\rangle$ sector H acts as $X \otimes X \otimes I$. Thus

$$|\Psi(t)\rangle = \frac{1}{\sqrt{2}}\Big((\cos t\,|000\rangle - i \sin t\,|110\rangle)|0\rangle + |000\rangle|1\rangle \Big).$$

The environment branches are orthogonal, so

$$\rho_\sigma(t) = \frac{1}{2}|\varphi(t)\rangle\langle\varphi(t)| + \frac{1}{2}|000\rangle\langle 000|,$$

where $|\varphi(t)\rangle = \cos t\,|000\rangle - i \sin t\,|110\rangle$.

Let $c = \cos t$, $s = \sin t$, $a = (1 + c^2)/2$, $b = s^2/2$, and $w = ics/2$. Then $a + b = 1$, and the only nonzero computational matrix elements of ρ_σ are

$$\langle 000|\rho_\sigma|000\rangle = a, \quad \langle 110|\rho_\sigma|110\rangle = b, \quad \langle 000|\rho_\sigma|110\rangle = w.$$

The site-3 marginal is $|0\rangle\langle 0|$. The site-1 and site-2 marginals are $\mathrm{diag}(a, b)$. Set $F = \mathrm{diag}(a, b) \otimes \mathrm{diag}(a, b) \otimes |0\rangle\langle 0|$. The difference $f = \rho_\sigma - F$ is zero off $\mathrm{span}\{|000\rangle, |100\rangle, |010\rangle, |110\rangle\}$. On that span its diagonal entries are $ab, -ab, -ab, ab$, and its only off-diagonal entry is w between $|000\rangle$ and $|110\rangle$. The eigenvalues are $-ab, -ab, ab + |w|, ab - |w|$. The singular values are $ab, ab, ab + |w|, |ab - |w||$, so

$$D_\sigma(t) = 2ab + 2\max(ab, |w|),$$

with $ab = s^2(1 + c^2)/4$ and $|w| = |cs|/2$.

On a right neighborhood of 0, $ab \le |w|$ and

$$D_\sigma(t) = |\sin t \cos t| + \frac{1}{2}\sin^2 t\,(1 + \cos^2 t).$$

The right derivative at 0 is 1. Since $f(0) = 0$, $\nu_\sigma = \|f'(0)\|_1 = 1$. Therefore $D_\sigma(t) \ge t - 10t^2$, and $D_\sigma(t) \ge t/2$ for $0 \le t \le 1/20$.

For $t \in [0, \pi/2]$ the nonzero eigenvalues of $\rho_\sigma(t)$ are $(1 \pm \cos t)/2$. They are not constant, so the reduced trajectory is not a unitary orbit on $\mathcal{H}_\sigma$. At $t = \pi/2$ the state is $\frac{1}{2}|000\rangle\langle 000| + \frac{1}{2}|110\rangle\langle 110|$, with purity $1/2$ and $D_\sigma = 1$.

This example is an exact symbolic witness. It is not a Python test. The code path for Theorem 4.3 is unchanged and still uses the weaker coefficient.

Lemma 4.3c (A jump trajectory with no finite autonomous Hermitian dilation)

Let $k = 2$ and

$$\mathcal{L}(\rho) = K\rho K^\dagger - \frac{1}{2}\{K^\dagger K, \rho\}, \qquad K = |00\rangle\langle 11|.$$

Set $\rho(0) = |11\rangle\langle 11|$. Then, for every $t \ge 0$,

$$\rho(t) = (1 - e^{-t})|00\rangle\langle 00| + e^{-t}|11\rangle\langle 11|.$$

Both single-site marginals equal $(1 - e^{-t})|0\rangle\langle 0| + e^{-t}|1\rangle\langle 1|$, and

$$D(t) = 4e^{-t}(1 - e^{-t}), \qquad \nu = \|f'(0)\|_1 = 4.$$

Hence $D(t) \geq 4t - 6t^2$. If $0 \leq t \leq 1/3$, then $D(t) \geq 2t$.

No fixed finite-dimensional time-independent Hermitian Hamiltonian, with a fixed initial total state, has this reduced trajectory for all $t \geq 0$. The obstruction is that the population $\langle 11|\rho(t)|11\rangle = e^{-t}$ is not a finite sum of characters $e^{-i\omega t}$. Lemma 4.3b therefore supplies neither a norm g nor site budgets g_j. The quadratic lower bound still holds for this one generator, with coefficient 6, by the scalar inequality in the proof. Replacing g by $\max_j g_j$ is not settled. Lemma 4.3d gives a coarser coefficient for a finite family of jumps and no Hamiltonian. It does not recover the coefficient 6. Lemma 4.3e puts a bounded Hamiltonian beside that family. The resulting coefficient is $2(1+k^2)(g+G)^2$, not 6.

Proof. Let $p(t) = \langle 11|\rho(t)|11\rangle$. On diagonal states the generator acts by $\mathcal{L}(|11\rangle\langle 11|) = |00\rangle\langle 00| - |11\rangle\langle 11|$ and $\mathcal{L}(|00\rangle\langle 00|) = 0$. Thus $\dot{p} = -p$ with $p(0) = 1$, so $p(t) = e^{-t}$ and $\rho(t) = (1 - e^{-t})|00\rangle\langle 00| + e^{-t}|11\rangle\langle 11|$. The partial traces are the stated marginals. Their product is diagonal, with entries $(1-e^{-t})^2$, $(1-e^{-t})e^{-t}$, $(1-e^{-t})e^{-t}$, and e^{-2t}. Subtracting from $\rho(t)$ leaves the diagonal

$$e^{-t}(1 - e^{-t}) \cdot (1, -1, -1, 1).$$

The trace norm is $D(t) = 4e^{-t}(1 - e^{-t})$. At $t = 0$ the derivative of this diagonal matrix is $\mathrm{diag}(1, -1, -1, 1)$, whose trace norm is 4. Since $f(0) = 0$, one has $\nu = 4$.

Set $u(t) = e^{-t} - e^{-2t}$, so $D = 4u$, $u(0) = 0$, and $u'(0) = 1$. Then

$$u''(t) + 3 = e^{-t} - 4e^{-2t} + 3.$$

With $s = e^{-t} \in (0, 1]$,

$$s - 4s^2 + 3 = -4(s - 1)(s + \tfrac{3}{4}).$$

The right-hand side is nonnegative on $(0, 1]$. Thus $u'' \geq -3$. The function $u'(t) - (1 - 3t)$ vanishes at 0 and has nonnegative derivative, so $u'(t) \geq 1 - 3t$. Integrating from 0 gives $u(t) \geq t - \frac{3}{2}t^2$. Therefore $D(t) \geq 4t - 6t^2$. On $0 \leq t \leq 1/3$ the quadratic term is at most $2t$, and $D(t) \geq 2t$.

Suppose a fixed finite-dimensional time-independent Hermitian H and a density ρ_{tot} satisfy $\tau(e^{-itH}\rho_{\mathrm{tot}}e^{itH}) = \rho(t)$ for every $t \geq 0$. In an eigenbasis of H, the population $\langle 11|\rho(t)|11\rangle$ would be a finite sum $\sum_k c_k e^{-i\omega_k t}$ with real frequencies ω_k. The functions $e^{\lambda t}$ for distinct complex λ are linearly independent on $[0, \infty)$: a shortest nontrivial relation $\sum_j c_j e^{\lambda_j t} = 0$, differentiated and subtracted against λ_n times itself, produces a shorter relation. The exponent -1 is not equal to any $-i\omega_k$. Independence forces the coefficient of e^{-t} to vanish, contradicting $\langle 11|\rho(t)|11\rangle = e^{-t}$. Hence no such finite autonomous Hermitian dilation exists. □

Lemma 4.3d (Quadratic coefficient for a finite family of jumps)

Let $k \geq 2$, and let the carrier be a finite product of k sites. Let

$$\mathcal{L}(X) = \sum_{\alpha=1}^{m} \Big(K_\alpha X K_\alpha^\dagger - \frac{1}{2}\{K_\alpha^\dagger K_\alpha,\, X\} \Big),$$

with no Hamiltonian term. Let

$$G \geq \sum_{\alpha=1}^{m} \|K_\alpha\|_\infty^2.$$

Let $\rho(0)$ be a product of its single-site marginals, and let $\rho(t) = \exp(t\mathcal{L})\rho(0)$. Keep D, f, and the single-site marginals of Theorem 4.3, and set $\nu = \|f'(0)\|_1$. Then, for every $t \geq 0$,

$$D(t) \geq \nu t - 2(1+k^2)G^2t^2 \geq \nu t - 6k^2G^2t^2.$$

If $\nu > 0$ and $0 \leq t \leq \nu/(4(1+k^2)G^2)$, then $D(t) \geq \nu t/2$. The weaker coefficient yields the same conclusion on the smaller window $0 \leq t \leq \nu/(12k^2G^2)$.

The bound is a uniform second-derivative estimate, not an exact defect formula. On the data of Lemma 4.3c one may take $m = 1$, $K_1 = |00\rangle\langle 11|$, and $G = 1$. Then $2(1+k^2)G^2 = 10$ and the half-speed window is $0 \leq t \leq 1/5$. Lemma 4.3c's coefficient is 6 and its window is $0 \leq t \leq 1/3$. Those sharper scalar values are not recovered. At $t = 0$, $\|\rho'(0)\|_1 = 2 = 2G$, so the first-derivative budget is achieved on that example. The quadratic coefficient is not.

No Hamiltonian summand is included in this lemma. Lemma 4.3e puts a bounded Hamiltonian beside the same jump family and replaces the quadratic coefficient by $2(1+k^2)(g+G)^2$. That extension does not recover the coefficient 6. The supports of the jumps are not used, and G is not replaced by a maximum over sites. The global factor g of Lemmas 4.3a and 4.3b is not removed, and replacement of g by $\max_j g_j$ is not claimed. The population argument of Lemma 4.3c is not extended: this lemma does not assert that a general jump trajectory lacks a finite autonomous Hermitian dilation.

Proof. For every matrix K, $\|K\|_\infty^2 = \|K^\dagger K\|_\infty$, because $\|K\|_\infty^2 = \sup_{\|v\|=1}\langle v, K^\dagger K v\rangle$ and $K^\dagger K$ is positive. For every matrix X and all bounded B, C,

$$\|BXC\|_1 \leq \|B\|_\infty \|X\|_1 \|C\|_\infty.$$

Indeed $\|X\|_1 = \sup_{\|W\|_\infty \leq 1} |\operatorname{Tr}(W^\dagger X)|$, and

$$\begin{aligned} |\operatorname{Tr}(W^\dagger BXC)| &= |\operatorname{Tr}((CW^\dagger B)X)| \\ &\leq \|CW^\dagger B\|_\infty \|X\|_1 \\ &\leq \|C\|_\infty \|W\|_\infty \|B\|_\infty \|X\|_1. \end{aligned}$$

If $\|W\|_\infty \leq 1$, the right-hand side is at most $\|B\|_\infty\|C\|_\infty\|X\|_1$. Taking the supremum gives the sandwich. Thus $\|KXK^\dagger\|_1 \leq \|K\|_\infty^2\|X\|_1$ and $\|K^\dagger KX\|_1 \leq \|K\|_\infty^2\|X\|_1$, and the same bound holds for $XK^\dagger K$. Each jump contributes at most $2\|K_\alpha\|_\infty^2\|X\|_1$, so

$$\|\mathcal{L}(X)\|_1 \leq 2G\|X\|_1$$

for every matrix X.

The generator $\mathcal{L}$ is trace-annihilating and Hermiticity-preserving. For the trace, $\operatorname{Tr}(KXK^\dagger) = \operatorname{Tr}(K^\dagger KX)$ and $\operatorname{Tr}(\{K^\dagger K, X\}) = 2\operatorname{Tr}(K^\dagger KX)$. For the adjoint, $(KXK^\dagger)^\dagger = KX^\dagger K^\dagger$ and $(K^\dagger KX)^\dagger = X^\dagger K^\dagger K$, so the formula for $\mathcal{L}(X^\dagger)$ is $\mathcal{L}(X)^\dagger$. The bound $\|\mathcal{L}(X)\|_1 \leq 2G\|X\|_1$ makes $\sum_n t^n/n!\,\mathcal{L}^n(\rho(0))$ absolutely convergent. Call the sum $\rho(t)$. Termwise differentiation, dominated by the same scalar majorant, gives $\rho'(t) = \mathcal{L}(\rho(t))$. For $n \geq 1$, $\mathcal{L}^n(\rho(0))$ is Hermitian and has trace zero, so $\rho(t)$ is Hermitian and $\operatorname{Tr}\rho(t) = 1$. The exponential semigroup $e^{t\mathcal{L}}$ preserves trace and Hermiticity.

Let $A = -\frac{1}{2}\sum_\alpha K_\alpha^\dagger K_\alpha$, and write $\mathcal{J}(X) = \sum_\alpha K_\alpha X K_\alpha^\dagger$, so $\mathcal{L}(X) = \mathcal{J}(X) + AX + XA$. The matrix A is Hermitian. Set $\Phi_t(X) = e^{tA}Xe^{tA}$. Then $\frac{d}{dt}\Phi_t(X) = A\Phi_t(X) + \Phi_t(X)A$. The same sandwich gives $\|\mathcal{J}(X)\|_1 \leq G\|X\|_1$. Also $\langle v, Av\rangle = -\frac{1}{2}\sum_\alpha \|K_\alpha v\|^2 \leq 0$. For $t \geq 0$ and any vector v,

$$\frac{d}{dt}\|e^{tA}v\|^2 = 2\langle e^{tA}v,\, Ae^{tA}v\rangle \leq 0,$$

so $\|e^{tA}\|_\infty \le 1$ and $\|\Phi_t(X)\|_1 \le \|X\|_1$.

If $\delta'(t) = \mathcal{L}(\delta(t))$ and $\delta(0) = 0$, the bound on $\mathcal{L}$ gives $\|\delta(t)\|_1 \le \int_0^t 2G\|\delta(s)\|_1\, ds$. Iterating and writing $M(t) = \sup_{0\le s\le t}\|\delta(s)\|_1$ produces $\|\delta(t)\|_1 \le M(t)(2Gt)^n/n!$ for every n, hence $\delta(t) = 0$. The linear equation therefore has at most one solution.

Define $\rho_0(t) = \Phi_t(\rho(0))$ and

$$\rho_{n+1}(t) = \int_0^t \Phi_{t-s}(\mathcal{J}(\rho_n(s)))\, ds.$$

The norm bounds give $\|\rho_0(t)\|_1 \le 1$. If $\|\rho_n(s)\|_1 \le (Gs)^n/n!$, then

$$\|\rho_{n+1}(t)\|_1 \le \int_0^t G\frac{(Gs)^n}{n!}\, ds = \frac{(Gt)^{n+1}}{(n+1)!}.$$

Thus $\sum_n \|\rho_n(t)\|_1 \le e^{Gt}$. Let $S(t) = \sum_{n=0}^\infty \rho_n(t)$. The majorant $\sum_n G(Gs)^n/n! = Ge^{Gs}$ is integrable on $[0,t]$, so

$$S(t) = \Phi_t(\rho(0)) + \int_0^t \Phi_{t-s}(\mathcal{J}(S(s)))\, ds.$$

The sum S is continuous, so the integrand is continuous for $0 \le s \le t$. Leibniz's rule differentiates $\Phi_t(\rho(0))$ into $A\Phi_t(\rho(0)) + \Phi_t(\rho(0))A$, adds the upper-limit term $\mathcal{J}(S(t))$, and adds

$$\int_0^t (A\Phi_{t-s}(\mathcal{J}(S(s))) + \Phi_{t-s}(\mathcal{J}(S(s)))A)\, ds.$$

Therefore $S'(t) = AS(t) + S(t)A + \mathcal{J}(S(t)) = \mathcal{L}(S(t))$, and $S(0) = \rho(0)$. Uniqueness gives $S = \rho$.

If X is positive semidefinite, then so are $\mathcal{J}(X)$ and $\Phi_t(X)$, the latter because $\langle v, \Phi_t(X)v\rangle = \langle e^{tA}v,\, Xe^{tA}v\rangle$. Each $\rho_n(t)$ is therefore positive semidefinite. The partial sums are positive semidefinite and that cone is closed, so the sum $\rho(t)$ is positive semidefinite. Its eigenvalues are nonnegative and sum to 1, hence $\|\rho(t)\|_1 = 1$.

Consequently $\|\rho'(t)\|_1 = \|\mathcal{L}(\rho(t))\|_1 \le 2G$. The same estimate on the matrix $\rho'(t)$ gives $\|\rho''(t)\|_1 \le 2G\|\rho'(t)\|_1 \le 4G^2$. Partial trace is trace-norm contractive on Hermitian inputs, and both derivatives are Hermitian, so $\|\rho_j'(t)\|_1 \le 2G$ and $\|\rho_j''(t)\|_1 \le 4G^2$. Every marginal density has trace norm one. Differentiating $F(t) = \bigotimes_j \rho_j(t)$ and using trace-norm multiplicativity, as in Theorem 4.3, yields

$$\begin{aligned}\|F''(t)\|_1 &\le \sum_j 4G^2 + \sum_{i\ne j}(2G)(2G)\\ &= 4kG^2 + 4k(k-1)G^2 = 4k^2G^2.\end{aligned}$$

Hence $\|f''(t)\|_1 \le 4G^2 + 4k^2G^2 = 4(1+k^2)G^2$.

Finite dimensionality makes $t \mapsto \rho(t)$ smooth. Since $\rho(0)$ is a product, $f(0) = 0$. As in Theorem 4.3,

$$f(t) = tf'(0) + \int_0^t (t-s)f''(s)\, ds,$$

so the remainder has trace norm at most $4(1+k^2)G^2 \cdot t^2/2 = 2(1+k^2)G^2t^2$. The reverse triangle inequality is the first bound. For $k \ge 2$,

$$6k^2 - 2(1+k^2) = 2(2k^2-1) \ge 14 > 0,$$

so the second bound is weaker.

If $\nu > 0$, then $G > 0$. The product rule and the bounds already proved give $\|F'(0)\|_1 \le \sum_j 2G = 2kG$, hence $\nu = \|f'(0)\|_1 \le 2G + 2kG = 2(1+k)G$. The stated window makes $2(1+k^2)G^2t^2$ at most $\nu t/2$. The weaker window does the same for $6k^2G^2t^2$.

On the Lemma 4.3c data, $G = 1$ and $k = 2$, so the quadratic coefficient is 10 and $\nu/(4(1+k^2)G^2) = 4/20 = 1/5$. Lemma 4.3c records $\mathcal{L}(|11\rangle\langle 11|) = |00\rangle\langle 00| - |11\rangle\langle 11|$, whose trace norm is $2 = 2G$. The comparison with the coefficient 6 is substitution of that lemma's bound. □

Lemma 4.3e (Quadratic coefficient for a bounded Hamiltonian beside jumps)

Let $k \ge 2$, and let the carrier be a finite product of k sites. Let $H = H^\dagger$ and

$$\mathcal{L}(X) = -i[H, X] + \sum_{\alpha=1}^{m} \Big(K_\alpha X K_\alpha^\dagger - \frac{1}{2}\{K_\alpha^\dagger K_\alpha,\, X\} \Big),$$

with $g \ge \|H\|_\infty$ and

$$G \ge \sum_{\alpha=1}^{m} \|K_\alpha\|_\infty^2.$$

Let $\rho(0)$ be a product of its single-site marginals, and let $\rho(t) = \exp(t\mathcal{L})\rho(0)$. Keep D, f, and the single-site marginals of Theorem 4.3, and set $\nu = \|f'(0)\|_1$. Then, for every $t \ge 0$,

$$D(t) \ge \nu t - 2(1+k^2)(g+G)^2 t^2 \ge \nu t - 6k^2(g+G)^2 t^2.$$

If $\nu > 0$ and $0 \le t \le \nu/(4(1+k^2)(g+G)^2)$, then $D(t) \ge \nu t/2$. The weaker coefficient yields the same conclusion on the smaller window $0 \le t \le \nu/(12k^2(g+G)^2)$.

If $G = 0$, the quadratic coefficient equals the leading coefficient $2(1+k^2)g^2$ of Theorem 4.3. If $g = 0$, it equals the coefficient $2(1+k^2)G^2$ of Lemma 4.3d.

The bound is a uniform second-derivative estimate. The triangle inequality produces the square $(g+G)^2 = g^2 + 2gG + G^2$. The cross term $2gG$ is not removed, and $(g+G)^2$ is not replaced by $g^2 + G^2$. Lemma 4.3f shows that the state-acceleration bound used to reach this square cannot make that replacement. The defect coefficient of this lemma is not improved. The scalar coefficient 6 and the window $1/3$ are not recovered.

On the witness $k = 2$, $m = 1$, $K = |00\rangle\langle 11|$, $H = |00\rangle\langle 00| - |11\rangle\langle 11|$, and $\rho(0) = |11\rangle\langle 11|$, one has $g = 1$ and $G = 1$. The quadratic coefficient is 40 and the half-speed window is $0 \le t \le 1/20$. The Hamiltonian commutes with the trajectory of Lemma 4.3c, so that trajectory is the mixed evolution. Lemma 4.3c still gives the coefficient 6 and the window $1/3$. At $t = 0$, $\|\rho'(0)\|_1 = 2$, while the first-derivative budget is $2(g+G) = 4$. The budget is not achieved, and the quadratic coefficient is not sharp.

Site budgets of H are not used. Jump supports are not used. The global factor g is not removed, and replacement by $\max_j g_j$ is not claimed. This lemma does not assert that a general mixed trajectory lacks a finite autonomous Hermitian dilation. The witness trajectory equals that of Lemma 4.3c, so that one trajectory still has none. The witness is exact and symbolic. It is not a Python test. The code path for Theorem 4.3 is unchanged.

Proof. The carrier is finite-dimensional, so $\mathcal{L}$ is a bounded linear map on a finite-dimensional space and the power series for $\exp(t\mathcal{L})$ converges absolutely. Write

$$A = -\frac{1}{2}\sum_{\alpha=1}^{m} K_\alpha^\dagger K_\alpha, \qquad \mathcal{J}(X) = \sum_{\alpha=1}^{m} K_\alpha X K_\alpha^\dagger,$$

and $B = -iH + A$. Then $A = A^\dagger$, $B^\dagger = iH + A$, and

$$B + B^\dagger = 2A = -\sum_{\alpha=1}^{m} K_\alpha^\dagger K_\alpha.$$

For every vector w, $\langle w, (B + B^\dagger)w\rangle = -\sum_\alpha \|K_\alpha w\|^2 \le 0$. For $t \ge 0$ and any vector v,

$$\frac{d}{dt}\|e^{tB}v\|^2 = \langle e^{tB}v,\, (B + B^\dagger)e^{tB}v\rangle \le 0,$$

so $\|e^{tB}\|_\infty \le 1$. The same identity for $B^\dagger$ gives $\|e^{tB^\dagger}\|_\infty \le 1$.

Lemma 4.3d's sandwich $\|CXD\|_1 \le \|C\|_\infty \|X\|_1 \|D\|_\infty$ therefore yields $\|\Phi_t(X)\|_1 \le \|X\|_1$ for $\Phi_t(X) = e^{tB}Xe^{tB^\dagger}$, and $\|\mathcal{J}(X)\|_1 \le G\|X\|_1$. The same lemma bounds the jump summand by $2G\|X\|_1$. The commutator estimate $\|[H, X]\|_1 \le 2\|H\|_\infty \|X\|_1$ gives

$$\|\mathcal{L}(X)\|_1 \le 2g\|X\|_1 + 2G\|X\|_1 = 2(g + G)\|X\|_1.$$

Trace: $\operatorname{Tr}(-i[H, X]) = 0$, and $\operatorname{Tr}(KXK^\dagger) = \operatorname{Tr}(K^\dagger KX)$ while $\operatorname{Tr}(\{K^\dagger K, X\}) = 2\operatorname{Tr}(K^\dagger KX)$, so each jump has trace zero. Adjoints: $(-i[H, X])^\dagger = -i[H, X^\dagger]$ because $H = H^\dagger$, and $(KXK^\dagger)^\dagger = KX^\dagger K^\dagger$, $(K^\dagger KX)^\dagger = X^\dagger K^\dagger K$. Thus $\mathcal{L}$ is trace-annihilating and Hermiticity-preserving. For $n \ge 1$, $\mathcal{L}^n(\rho(0))$ is Hermitian and has trace zero, so $\rho(t)$ is Hermitian and $\operatorname{Tr}\rho(t) = 1$. The exponential semigroup $e^{t\mathcal{L}}$ preserves trace and Hermiticity. Termwise differentiation of the power series gives $\rho'(t) = \mathcal{L}(\rho(t))$.

If $\delta'(t) = \mathcal{L}(\delta(t))$ and $\delta(0) = 0$, then $\|\delta(t)\|_1 \le \int_0^t 2(g + G)\|\delta(s)\|_1\, ds$. Writing $M(t) = \sup_{0\le s\le t}\|\delta(s)\|_1$ and iterating gives $\|\delta(t)\|_1 \le M(t)\,(2(g + G)t)^n/n!$ for every n, hence $\delta(t) = 0$. The linear equation has at most one solution.

Differentiate Φ_t. Since $\frac{d}{dt}e^{tB} = Be^{tB}$ and $\frac{d}{dt}e^{tB^\dagger} = e^{tB^\dagger}B^\dagger$,

$$\frac{d}{dt}\Phi_t(X) = B\Phi_t(X) + \Phi_t(X)B^\dagger = -i[H, \Phi_t(X)] + A\Phi_t(X) + \Phi_t(X)A.$$

Define $\rho_0(t) = \Phi_t(\rho(0))$ and

$$\rho_{n+1}(t) = \int_0^t \Phi_{t-s}(\mathcal{J}(\rho_n(s)))\, ds.$$

The norm bounds give $\|\rho_0(t)\|_1 \le 1$. If $\|\rho_n(s)\|_1 \le (Gs)^n/n!$, then

$$\|\rho_{n+1}(t)\|_1 \le \int_0^t G\frac{(Gs)^n}{n!}\, ds = \frac{(Gt)^{n+1}}{(n+1)!}.$$

Thus $\sum_n \|\rho_n(t)\|_1 \le e^{Gt}$. Let $S(t) = \sum_{n=0}^\infty \rho_n(t)$. The majorant $\sum_n G(Gs)^n/n! = Ge^{Gs}$ is integrable on $[0, t]$, so

$$S(t) = \Phi_t(\rho(0)) + \int_0^t \Phi_{t-s}(\mathcal{J}(S(s)))\, ds.$$

The sum S is continuous, so the integrand is continuous for $0 \le s \le t$. Leibniz's rule differentiates $\Phi_t(\rho(0))$ into $-i[H, \Phi_t(\rho(0))] + A\Phi_t(\rho(0)) + \Phi_t(\rho(0))A$, adds the upper-limit term $\mathcal{J}(S(t))$, and adds

$$\int_0^t \left(-i[H, \Phi_{t-s}(\mathcal{J}(S(s)))] + A\Phi_{t-s}(\mathcal{J}(S(s))) + \Phi_{t-s}(\mathcal{J}(S(s)))A\right) ds.$$

Therefore $S'(t) = \mathcal{L}(S(t))$ and $S(0) = \rho(0)$. Uniqueness gives $S = \rho$.

If X is positive semidefinite, then so is $\mathcal{J}(X)$, and $\langle v, \Phi_t(X)v\rangle = \langle e^{tB^\dagger}v,\, Xe^{tB^\dagger}v\rangle \ge 0$. Each $\rho_n(t)$ is therefore positive semidefinite. The partial sums are positive semidefinite and that cone is closed, so $\rho(t)$ is positive semidefinite. Its eigenvalues are nonnegative and sum to 1, hence $\|\rho(t)\|_1 = 1$.

Consequently $\|\rho'(t)\|_1 = \|\mathcal{L}(\rho(t))\|_1 \le 2(g+G)$. The same estimate on $\rho'(t)$ gives $\|\rho''(t)\|_1 \le 2(g+G)\|\rho'(t)\|_1 \le 4(g+G)^2$. Both derivatives are Hermitian. Partial trace is trace-norm contractive on Hermitian inputs, so $\|\rho_j'(t)\|_1 \le 2(g+G)$ and $\|\rho_j''(t)\|_1 \le 4(g+G)^2$. Every marginal density has trace norm one. Differentiating $F(t) = \bigotimes_j \rho_j(t)$ and using trace-norm multiplicativity, as in Theorem 4.3, yields

$$\begin{aligned}\|F''(t)\|_1 &\le \sum_j 4(g+G)^2 + \sum_{i\ne j}\big(2(g+G)\big)^2 \\ &= 4k(g+G)^2 + 4k(k-1)(g+G)^2 = 4k^2(g+G)^2.\end{aligned}$$

Hence $\|f''(t)\|_1 \le 4(g+G)^2 + 4k^2(g+G)^2 = 4(1+k^2)(g+G)^2$.

Since $\rho(0)$ is a product, $f(0) = 0$. As in Theorem 4.3,

$$f(t) = tf'(0) + \int_0^t (t-s)f''(s)\, ds,$$

so the remainder has trace norm at most $4(1+k^2)(g+G)^2 \cdot t^2/2 = 2(1+k^2)(g+G)^2t^2$. The reverse triangle inequality is the first bound. For $k \ge 2$,

$$6k^2 - 2(1+k^2) = 2(2k^2-1) \ge 14 > 0,$$

so the second bound is weaker.

If $\nu > 0$, then $g + G > 0$. The product rule and the bounds already proved give $\|F'(0)\|_1 \le \sum_j 2(g+G) = 2k(g+G)$, hence $\nu = \|f'(0)\|_1 \le 2(g+G) + 2k(g+G) = 2(1+k)(g+G)$. The stated window makes $2(1+k^2)(g+G)^2t^2$ at most $\nu t/2$. The weaker window does the same for $6k^2(g+G)^2t^2$.

The substitutions $G = 0$ and $g = 0$ are the two endpoint claims.

For the witness, order the basis as $|00\rangle, |01\rangle, |10\rangle, |11\rangle$. Then $H = \mathrm{diag}(1, 0, 0, -1)$, so $\|H\|_\infty = 1$. Also $\|K\|_\infty = 1$, and one may take $G = 1$. Let

$$\rho_\star(t) = (1 - e^{-t})|00\rangle\langle 00| + e^{-t}|11\rangle\langle 11|$$

be the trajectory of Lemma 4.3c. Both H and $\rho_\star(t)$ are diagonal in this basis, so $[H, \rho_\star(t)] = 0$. Lemma 4.3c records $\mathcal{L}_0(|11\rangle\langle 11|) = |00\rangle\langle 00| - |11\rangle\langle 11|$ and $\mathcal{L}_0(|00\rangle\langle 00|) = 0$ for the pure jump generator $\mathcal{L}_0$. Linearity gives

$$\mathcal{L}_0(\rho_\star(t)) = e^{-t}|00\rangle\langle 00| - e^{-t}|11\rangle\langle 11| = \rho_\star'(t).$$

Thus $\rho_\star'(t) = \mathcal{L}(\rho_\star(t))$ and $\rho_\star(0) = \rho(0)$. Uniqueness gives $\rho = \rho_\star$. Lemma 4.3c therefore supplies $D(t) = 4e^{-t}(1 - e^{-t})$, $\nu = 4$, the coefficient 6, and the window 1/3.

The mixed coefficient is $2(1 + 4)(1 + 1)^2 = 40$, and

$$\nu/(4(1 + k^2)(g + G)^2) = 4/(4 \cdot 5 \cdot 4) = 1/20.$$

At $t = 0$, $\rho'(0) = |00\rangle\langle 00| - |11\rangle\langle 11|$. Its eigenvalues are $1, 0, 0, -1$, so its trace norm is 2. The first-derivative budget is $2(g + G) = 4$. The population identity of Lemma 4.3c applies to $\rho_\star$, so this one trajectory has no finite autonomous Hermitian dilation. □

Lemma 4.3f (The acceleration cross term does not drop)

Let the carrier be two qubits. Let $S = |0\rangle\langle 0| + i|1\rangle\langle 1|$, $Z = |0\rangle\langle 0| - |1\rangle\langle 1|$, $H = Z \otimes I$, and $K = S \otimes I$. Let

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \qquad \rho(0) = |+\rangle\langle +| \otimes |0\rangle\langle 0|.$$

Then $g = \|H\|_\infty = 1$ and $G = \|K\|_\infty^2 = 1$ are admissible in Lemma 4.3e, and

$$\|\rho''(0)\|_1 = 10.$$

In particular

$$4(g^2 + G^2) = 8 < \|\rho''(0)\|_1 < 16 = 4(g + G)^2.$$

The state-acceleration estimate $\|\rho''(t)\|_1 \le 4(g+G)^2$ of that lemma cannot be replaced by $\|\rho''(t)\|_1 \le 4(g^2 + G^2)$.

The same data give $\|\rho'(0)\|_1 = \sqrt{10}$. The first-derivative budget $2(g + G) = 4$ is not achieved. The acceleration budget 16 is not achieved either. The witness forces a gap of 2 above $4(g^2 + G^2)$. It does not force the full cross contribution $8gG = 8$ inside $4(g + G)^2$.

The orbit is the product $A(t) \otimes |0\rangle\langle 0|$, with

$$A(t) = \begin{pmatrix} 1/2 & c(t) \\ \overline{c(t)} & 1/2 \end{pmatrix}, \qquad c(t) = \tfrac{1}{2}e^{-(1+3i)t}.$$

Thus $D(t) = 0$ and $\nu = 0$. Both $\nu t - 2(1+k^2)(g+G)^2t^2$ and $\nu t - 2(1+k^2)(g^2+G^2)t^2$ are nonpositive, so the witness does not test either defect coefficient. It does not replace $(g+G)^2$ by g^2+G^2 in the defect bound of Lemma 4.3e, and it does not establish that this defect replacement is false. The scalar coefficient 6 and the window 1/3 are not recovered. No site budgets are used. The factor e^{-t} in $c(t)$ is not a finite sum of characters $e^{-i\omega t}$, so this one trajectory has no finite autonomous Hermitian dilation. That is not a statement about a general mixed generator. The witness is exact and symbolic. It is not a Python test. The code path for Theorem 4.3 is unchanged.

Proof. The eigenvalues of S are 1 and i, so $\|S\|_\infty = 1$. Also $S^\dagger S = I$ and $\|Z\|_\infty = 1$. Hence $\|H\|_\infty = 1$ and $\|K\|_\infty = 1$.

Let $P = |0\rangle\langle 0|$ on the second factor. For every one-site matrix M,

$$[H, M \otimes P] = [Z, M] \otimes P,$$

$$K(M \otimes P)K^\dagger = (SMS^\dagger) \otimes P,$$

and $K^\dagger K = I$, so

$$\mathcal{L}(M \otimes P) = \mathcal{L}_1(M) \otimes P,$$

where $\mathcal{L}_1(M) = -i[Z, M] + SMS^\dagger - M$.

The diagonal entries of Z are 1 and -1, so $[Z, M]_{00} = [Z, M]_{11} = 0$. Since S is diagonal, $(SMS^\dagger)_{00} = M_{00}$ and $(SMS^\dagger)_{11} = M_{11}$. Thus $\mathcal{L}_1$ has zero diagonal. For the coherence $c = M_{01}$,

$$(-i[Z, M])_{01} = -2ic, \qquad (SMS^\dagger)_{01} = -ic,$$

because $(S^\dagger)_{11} = -i$. The $-M$ term contributes $-c$, so

$$\mathcal{L}_1(M)_{01} = -(2i + i + 1)c = -(1 + 3i)c.$$

Starting from $A(0) = |+\rangle\langle+|$, whose coherence is $1/2$ and whose diagonal is $(1/2, 1/2)$, the coherence solves $\dot{c} = -(1+3i)c$. Hence $c(t) = \frac{1}{2}e^{-(1+3i)t}$ and the diagonal of $A(t)$ stays $(1/2, 1/2)$. The matrix $A(t)$ is positive semidefinite because its eigenvalues are $1/2 \pm |c(t)|$ and $|c(t)| = e^{-t}/2 \leq 1/2$. It has trace one.

The product curve $\rho(t) = A(t) \otimes P$ is Hermitian, has trace one, and is positive semidefinite. The intertwining identity gives $\rho'(t) = \mathcal{L}(\rho(t))$, and $\rho(0)$ matches the initial state. Lemma 4.3e records uniqueness for this linear equation, so this curve is the mixed evolution. It is a product for every $t \geq 0$, and therefore $D(t) = 0$. Differentiating the product at $t = 0$ gives $f'(0) = 0$, so $\nu = 0$.

Differentiate the coherence twice:

$$\ddot{c}(0) = (1 + 3i)^2 \cdot \tfrac{1}{2} = \tfrac{1+6i-9}{2} = -4 + 3i.$$

Thus

$$\mathcal{L}_1^2(A(0)) = \begin{pmatrix} 0 & -4 + 3i \\ -4 - 3i & 0 \end{pmatrix}.$$

A Hermitian matrix $\begin{pmatrix} 0 & z \\ \bar{z} & 0 \end{pmatrix}$ has eigenvalues $\pm|z|$. Here $|-4 + 3i| = \sqrt{16 + 9} = 5$, so the trace norm is 10. Trace norm is multiplicative and $\|P\|_1 = 1$, hence $\|\rho''(0)\|_1 = 10$.

The separated value is $4(g^2 + G^2) = 8$. The Lemma 4.3e value is $4(g + G)^2 = 16$. The inequality $\|\rho''(0)\|_1 \leq 4(g^2 + G^2)$ is false, and $10 \leq 16$.

The first derivative has coherence $\dot{c}(0) = -(1 + 3i)/2$, so $\|\rho'(0)\|_1 = |1 + 3i| = \sqrt{10}$. The budget $2(g + G) = 4$ is larger.

Finally, $\langle 00|\rho(t)|10\rangle = c(t) = \frac{1}{2}e^{(-1-3i)t}$. If $\rho(t)$ were the reduction of a fixed finite-dimensional autonomous Hermitian evolution with a fixed initial total state, this entry would be a finite sum of characters $e^{-i\omega t}$, by the same spectral expansion used in Lemma 4.3c. The functions $e^{\lambda t}$ for distinct complex λ are linearly independent on $[0, \infty)$, again as in that lemma. The exponent $-1 - 3i$ is not purely imaginary, so $c(t)$ is not such a sum. This one trajectory has no finite autonomous Hermitian dilation. □

4.1 What this distance detects

For two parties this diagnostic is Hall's correlation distance, with the same trace-norm normalization Hall, 2013. Nonzero distance alone is compatible with separability. For two qubits, $D > 1$ is nevertheless a sufficient entanglement criterion; it is not necessary, and no corresponding multipartite criterion is asserted here.

The chosen marginal-preserving retraction is generally not a trace-norm nearest-product projection, as also discussed by Aaronson et al., 2013. For example, take $\rho = \mathrm{diag}(1/2, 0, 0, 1/2)$. Then $\Pi(\rho) = I_4/4$ and $D(\rho) = 1$. But the product state $\eta = \mathrm{diag}(0.7, 0.3) \otimes \mathrm{diag}(0.7, 0.3)$ has $\|\rho - \eta\|_1 = 0.01 + 0.21 + 0.21 + 0.41 = 0.84$. Thus D measures the residual of this specified retraction, rather than the infimum over all product states.

$D = 0$ means the state is the product of its own marginals. $D > 0$ means it is not fully factorized. A separable mixture can have $D > 0$: for

$$S = \tfrac{1}{2}|00\rangle\langle 00| + \tfrac{1}{2}|11\rangle\langle 11|,$$

$\Pi(S) = I_4/4$ and $D(S) = 1$, despite separability. Neither D, Q, nor R_k alone certifies genuine multipartite entanglement.

A zero initial speed is only a first-order statement. For $H = X \otimes I + Z \otimes X$ and $\rho = |00\rangle\langle 00|$, the initial velocity is tangent, hence $\nu = 0$. But $H^2 = 2I$, and

$$|\psi(t)\rangle = \cos(\sqrt{2}t)|00\rangle - \frac{i \sin(\sqrt{2}t)}{\sqrt{2}}(|10\rangle + |01\rangle).$$

Its two-by-two amplitude determinant is $\sin^2(\sqrt{2}t)/2$, nonzero at sufficiently small $t > 0$. Thus factorization can fail at second order.

Theorem 4.4 (Trace-norm polar witness)

For any matrix n, let its thin singular-value decomposition on nonzero singular directions be $n = U_r \Sigma_r V_r^\dagger$. Then

$$W = U_r V_r^\dagger, \qquad \|W\|_\infty \le 1, \qquad \mathrm{Re}\,\mathrm{Tr}(W^\dagger n) = \|n\|_1.$$

For $n = 0$, take $W = 0$.

Proof. W is a partial isometry, and its pairing with n is $\mathrm{Tr}\,\Sigma_r = \sum_i s_i$. Trace/operator norm duality bounds the pairing of any operator-norm contraction with n by $\|n\|_1$. Hence W is optimal. □

This witnesses the norm of a specified residual. It does not imply that W annihilates the product tangent space, or that the oblique residual is the nearest-point distance to that space.

Counterexample 4.4a (Polar optimality without tangent annihilation)

Take

$$\rho_A = \mathrm{diag}(3/4, 1/4), \quad \rho_B = \mathrm{diag}(1/3, 2/3), \quad n = \mathrm{diag}(1, -1, -1, 1).$$

Both marginals of n vanish, so $Pn = 0$. Its polar witness is $W = n$, with $\|W\|_\infty = 1$ and $\mathrm{Tr}(W^\dagger n) = 4$. But the tangent vector $Y = \mathrm{diag}(1, -1) \otimes \rho_B$ satisfies

$$\mathrm{Tr}(W^\dagger Y) = -2/3.$$

A tangent-annihilating dual would require an additional annihilator constraint. The polar construction does not supply it.

4.2 Conditional Duhamel observable calculation

For a differentiable approximation $m(t)$ with the same initial state as the exact trajectory, define $r(t) = -i[H, m(t)] - m'(t)$. If exact propagation is unitary, variation of constants gives

$$|\operatorname{Tr}(O^\dagger(\rho(t) - m(t)))| \le \|O\|_\infty \int_0^t \|r(s)\|_1 \, ds.$$

The software multiplies a **supplied** residual integral by $\|O\|_\infty$; it does not calculate or certify that integral from an initial speed.

4.3 Robust product defects and observed cochains (PO034-C8)

Proposition 4.5. Let ρ, σ be density operators on the same finite tensor product of $k \ge 1$ factors. Write $\Pi(\rho) = \bigotimes_{j=1}^k \rho_j$ and $D(\rho) = \|\rho - \Pi(\rho)\|_1$. Then

$$\|\Pi(\rho) - \Pi(\sigma)\|_1 \le k\|\rho - \sigma\|_1, \qquad |D(\rho) - D(\sigma)| \le (k+1)\|\rho - \sigma\|_1.$$

Proof. Telescope the product difference by replacing one marginal at a time. Every unchanged density factor has trace norm one, so the tensor-product norm identity bounds the result by $\sum_j \|\rho_j - \sigma_j\|_1$. Contractivity of partial trace on Hermitian inputs bounds each summand by $\|\rho - \sigma\|_1$. The reverse triangle inequality applied to the two residuals proves the second bound. No optimality of the factor $k+1$ is claimed. □

Suppose a validated density reference σ obeys $\|\rho - \sigma\|_1 \le \epsilon$, and a computed estimate obeys $|\widehat{D} - D(\sigma)| \le \eta_D$. Then

$$\max\{0, \widehat{D} - (k+1)\epsilon - \eta_D\} \le D(\rho) \le \widehat{D} + (k+1)\epsilon + \eta_D.$$

At $k = 2$, $\widehat{D} = 1/50$, $\epsilon = 1/150$ and $\eta_D = 0$, the guaranteed lower margin is zero. Reducing the state radius to $1/300$ raises that margin to $1/100$. A nonphysical floating-point matrix cannot serve as the density reference without a separately bounded repair to a density operator.

For a declared bounded linear observation A with $\|AX\| \le L\|X\|_1$, the same proof gives

$$\|A(\rho - \Pi\rho) - A(\sigma - \Pi\sigma)\| \le L(k+1)\epsilon.$$

The observation codomain and norm must be fixed. If it is a finite normed cochain space and M is the declared linear subspace of repairable cochains, the quotient-distance function $q(c) = \inf_{m \in M} \|c - m\|$ is 1-Lipschitz: the triangle inequality gives $q(c) \le \|c - d\| + q(d)$ and the reversed bound. Consequently the same uncertainty radius controls the observed quotient mass. This does not identify the quantum residual with a coboundary; §5's observation map remains a required input. Nor does a positive product defect alone establish entanglement.

The preceding estimate complements PO-033's correction-jump budget. A replacement of the approximate state by distance at most j_k consumes that amount in its trace-norm error ledger; the resulting product-defect uncertainty is at most $(k+1)j_k$. Exact admission of a rational Choi matrix and a numerical unitarity diagnostic have different meanings. Neither supplies a bound on an entire numerical trajectory without additional analysis.

The analytic inequalities are written proofs, not new Lean declarations. `code/residual_error_budget.py` and its tests evaluate the resulting scalar intervals with exact fractions. The valid and near-miss records in `docs/residual_error_budget/` check supplied bounds only. The trace-norm facts used in the proof are standard; see Watrous, Chapters 1–2.

5. From a residual to a cochain

Let K be a declared finite simplicial complex, with every face included and each simplex oriented by increasing vertex order. For each $\sigma \in K_p$, declare a coefficient space and a linear observation

$$\mathcal{O}_{\sigma,\rho_\sigma} : \ker P_{\rho_\sigma} \longrightarrow V_\sigma.$$

For scalar cochains, a concrete choice is a fixed Hermitian operator B_σ and $\mathcal{O}_\sigma(Z) = \mathrm{Re}\,\mathrm{Tr}(B_\sigma^\dagger Z)$. Assign $\omega(\sigma) = \mathcal{O}_\sigma(n_\sigma)$, with a sign change under orientation reversal. The fixed observation, orientation, carrier, and coefficient convention are all part of the data.

The residual norm is nonlinear and cannot silently replace this linear observation map. Independently chosen coefficient spaces also require restriction/transport maps before a sheaf coboundary can be formed. The current implementation uses ordinary scalar cochains with constant coefficients, not a general cellular sheaf.

For these scalar cochains,

$$(d_p\omega)(v_0,\dots,v_{p+1}) = \sum_{i=0}^{p+1}(-1)^i\omega(v_0,\dots,\widehat{v}_i,\dots,v_{p+1}), \qquad d_{p+1}d_p = 0.$$

Theorem 5.2 (Finite scalar Hodge decomposition)

Equip the oriented simplex basis with the Euclidean inner product and set

$$\Delta_p = d_{p-1}d_{p-1}^* + d_p^*d_p.$$

Then

$$C^p(K) = \mathrm{im}\,d_{p-1} \oplus^\perp \mathrm{im}\,d_p^* \oplus^\perp \ker\Delta_p.$$

Each cochain has unique **components** in these three spaces; potentials α, β in $\omega = d_{p-1}\alpha + d_p^*\beta + \gamma$ need not be unique.

Proof. The two image spaces are orthogonal because $d_pd_{p-1} = 0$. Also,

$$\langle\gamma, \Delta_p\gamma\rangle = \|d_{p-1}^*\gamma\|^2 + \|d_p\gamma\|^2.$$

Thus the kernel is their common orthogonal complement, and the finite orthogonal direct-sum decomposition follows. Harmonic vectors represent cohomology classes for this declared complex and coefficient system. □

For the hollow triangle with edge order $(01), (02), (12)$, the cycle cochain is $(1, -1, 1)$ and $\dim\ker\Delta_1 = 1$. Filling the triangle gives $\Delta_1 = 3I_3$, so this harmonic space disappears. Neither result establishes topological entanglement or immunity to product retraction. Those interpretations need a proved observation/dynamics bridge, which is not supplied here.

6. Graded squares and a separate semigroup defect

6.1 A consistent graded-square convention

Declare linear maps of bidegrees $(1, 0)$ and $(0, 1)$:

$$d_h^{p,q} : C^{p,q} \to C^{p+1,q}, \qquad d_v^{p,q} : C^{p,q} \to C^{p,q+1}.$$

Define the square defect by

$$C^{p,q} = d_h^{p,q+1} d_v^{p,q} - d_v^{p+1,q} d_h^{p,q}.$$

For totalization $D|_{C^{p,q}} = d_h + (-1)^p d_v$, direct expansion gives

$$D^2|_{C^{p,q}} = d_h^2 + d_v^2 + (-1)^p C^{p,q}.$$

If $d_h^2 = d_v^2 = 0$, then $D^2 = (-1)^p C$. Commuting horizontal and vertical maps yield a double complex and $D^2 = 0$. A nonzero square defect violates that double-complex condition.

The demonstration class alternates identity and zero vertical maps, so vertical nilpotence really holds; its square defect is zero. A separate noncommuting square tests the sign with $d_h = (1, -1)$, $d_v^{0,0} = I_2$, and $d_v^{1,0} = 2I_1$, giving $C = -(1, -1)$. These finite matrix constructions have no supplied Lie algebra, non-Abelian connection, or physical memory law.

Proposition 6.2 (State-local semigroup defect coefficient)

Independently of a cochain complex, let Π be a C^2 retraction on a finite ambient vector space, let A be linear, and let m be a fixed point of Π. Define

$$P = D\Pi_m, \quad a = Am, \quad v = Pa, \quad n = a - v, \quad F_t = \Pi \circ e^{tA}|_{\operatorname{im}\Pi}.$$

The mixed coefficient of $F_{s+t}(m) - F_s(F_t(m))$ at $s = t = 0$ is

$$\mathcal{K}_{\Pi,A}(m) = PAn + D^2\Pi_m(a, n).$$

Proof. The mixed derivative of $F_{s+t}(m)$ is $PA^2m + D^2\Pi_m(a, a)$. For $F_s(F_t(m))$, it is $PAv + D^2\Pi_m(v, a)$. Subtract and use symmetry and bilinearity of the Hessian. □

The coefficient is tangent for a genuine retraction. Differentiating $\Pi \circ \Pi = \Pi$ twice on (a, n) gives

$$D^2\Pi_m(Pa, Pn) + PD^2\Pi_m(a, n) = D^2\Pi_m(a, n).$$

Since $Pn = 0$, the Hessian term is fixed by P, as is PAn. For the product-of-marginals map on trace-zero directions, every marginal of n is zero, so $D^2\Pi_m(a, n) = 0$.

The numerical helper assembles the coefficient from supplied P, A, n and a supplied Hessian vector, checks the projector/normal conditions, and reports whether the result is tangent. It does not verify that an arbitrary input vector is a genuine Hessian. No equation identifies $\mathcal{K}_{\Pi,A}$ with the graded-square defect C.

7. A local protocol scaffold with declared evidence

The names Prime -1, 000, 001, and 002 are used here as a local organizing analogy for primitive difference, declaration, loss accounting, and a stop policy. These helpers do not implement or formally establish an external Prime algorithm.

7.1 Carrier declaration and adapter loss

A carrier records a simplex, its simplicial degree $p = |\sigma| - 1$, a residual norm, a budget, and identifiers for boundary, observation, and witness evidence. "Declaration valid" means that required metadata is present; the identifiers do not replay the maps or their proofs. Q is a complementary projector, not an automatically supplied quotient or boundary operator.

To compare face defects with a cell defect, first declare lifts E_f into a common operator space:

$$L_{\text{adapter}} = \left\| Z_\sigma - \sum_f E_f Z_f \right\|_1 .$$

The code accepts the already lifted matrices and checks their shapes. Subtracting matrices from different Hilbert spaces is undefined. The scalar difference $\left| \|Z_\sigma\|_1 - \| \sum_f E_f Z_f \|_1 \right|$ is only a lower bound. For a single lifted defect $-Z_\sigma$, that difference is zero while the actual loss is $2\|Z_\sigma\|_1$.

7.2 Stop rules are policies

The helper can stop for zero initial speed, a degree cap, small declared adapter loss, depleted budget, repeated supplied full-state fingerprint, or absent witness evidence. A complete declaration is also required. These decisions do not establish exact future dynamics or certify ascent. Equal scalar residual norms cannot establish a repeated carrier state. Passing the policy permits the next check; it grants no authority.

7.3 Seven-step trace schema

This paper's local sequence is

$$[E, R, C, \Lambda, D, M, T],$$

which contains **seven** labels. It is not claimed to be an eight-letter DDC alphabet. The labels request state declaration, boundary declaration, retraction, Laplacian evaluation, evolution, gluing checks, and a receipt. The scaffold does not perform those numerical operations and always sets `receipt_verified = False`. It supplies no invented decay factors, harmonic estimates, or "glued" status.

8. Worked examples and evidence boundaries

8.1 Three-qubit GHZ-family evolution on a degree-2 simplex

Let $\rho = |000\rangle\langle 000|$, $H = gX_1X_2X_3$, and $g > 0$. Then

$$L = -ig(|111\rangle\langle 000| - |000\rangle\langle 111|), \quad PL = 0, \quad QL = R_3 L = L, \quad \nu = 2g.$$

The exact pure state is

$$|\psi(t)\rangle = \cos(gt)|000\rangle - i\sin(gt)|111\rangle.$$

Theorem 4.3 gives

$$D(t) \geq 2gt - 54g^2t^2 \quad \text{and} \quad D(t) \geq gt \quad (0 \leq t \leq 1/(54g)).$$

This Hamiltonian is one full-support term, so every site budget equals g. Lemma 4.3a then reproduces the leading coefficient $20g^2$, not the weaker $54g^2$ displayed above. The notebook constructs this analytic state, computes its trace norm numerically, and compares it with the bound at declared times. It also computes Q and R_3 on the two-site counterexample.

The companion figure plots this finite example and the lower bound. It is not empirical convergence data.

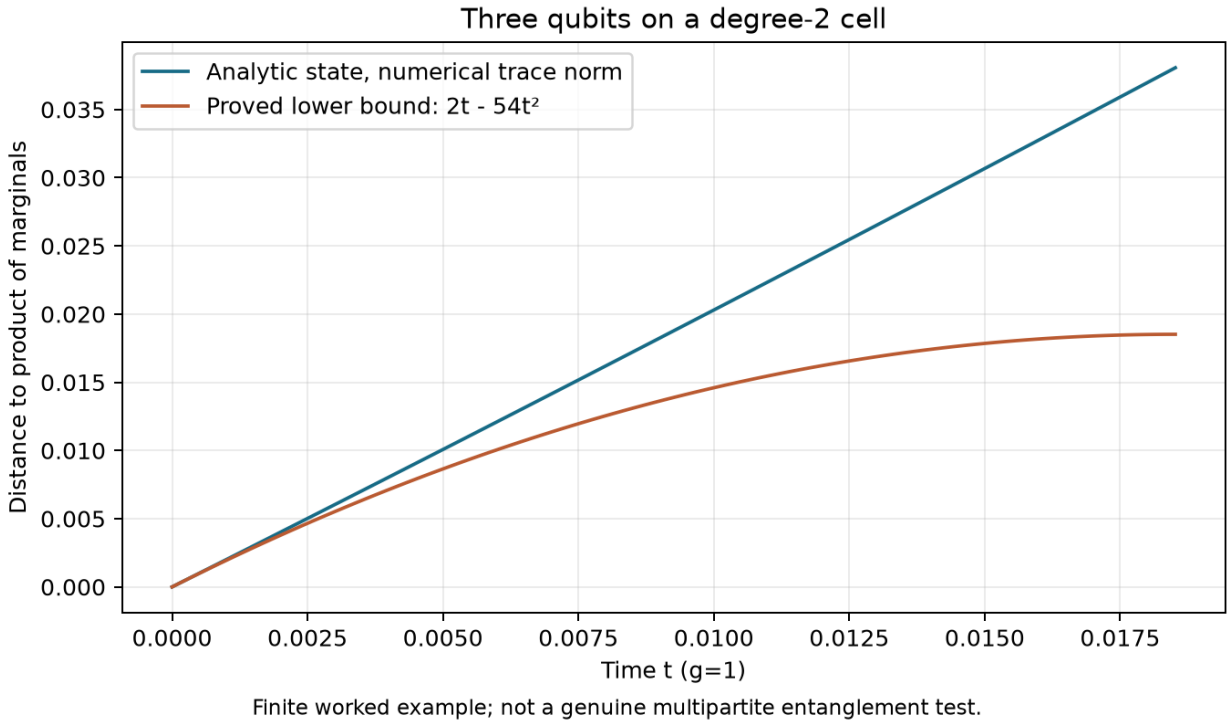


Figure 1: Three-qubit analytic-state defect and the proved lower bound for g=1.

8.2 Evidence inventory

Result	Evidence in this package	Limit
Projector algebra	Theorem 4.1; abstract Lean declarations; numerical tests	Lean assumes trace/embedding identities as structure fields
One-site isolation	Theorem 4.2; abstract one-site Lean declarations	No formal proof of all proper-support isolation
Highest-order isolation	Proposition 4.2b; numerical regressions	Separate projector, not a formal density-matrix instantiation
Short-time bound	Theorem 4.3; analytic-state comparisons	Entire-Hamiltonian norm hypothesis; code uses the weaker coefficient; not mechanized
Site-budget refinement	Lemma 4.3a; exact symbolic witness	Global norm remains in the coefficient; closed system; not mechanized
Reduced-subsystem bound	Lemma 4.3b; exact symbolic witness	Dilation norm remains; not a general Lindblad theorem; not mechanized
Jump trajectory without a finite dilation	Lemma 4.3c; exact scalar witness	One generator only; scalar coefficient 6; not mechanized
Finite jump family	Lemma 4.3d; comparison with Lemma 4.3c	Coefficient $2(1+k^2)G^2$; no Hamiltonian; no site budgets; 10 rather than 6 on Lemma 4.3c; not mechanized

Result	Evidence in this package	Limit
Hamiltonian beside jumps	Lemma 4.3e; commuting witness	Coefficient $2(1+k^2)(g+G)^2$; 40 rather than 6 on the witness; cross term kept; no site budgets; not mechanized
Acceleration cross term	Lemma 4.3f; exact product witness	$\|\rho''(0)\|_1 = 10 > 8 = 4(g^2+G^2)$; defect coefficient unchanged; $\nu = 0$; not mechanized
Polar witness	Theorem 4.4; numerical SVD	No automatic tangent-annihilator property
Hodge decomposition	Theorem 5.2; scalar incidence matrices	No sheaf or quantum-entanglement bridge
Square / semigroup defects	Separate derivations and regression examples	No automatic identification between the two
Protocol scaffold	Metadata and policy checks	No executed DDC simulation or authority

The Lean package is pinned to Lean 4.34.1. Its seven named declarations establish consequences of the supplied abstract trace/embedding laws. The axiom report's "no axioms" output means no global logical axioms appear in those declarations; it does **not** remove their explicit structure hypotheses. It is a stored report, not a fresh build receipt for this review. The concrete density-matrix instantiation remains unmechanized.

The Python core uses NumPy floating-point linear algebra. It is not an exact rational certifier, and there is no Rust implementation in this deposit.

The review receipt records executed tests, notebook execution method, rendering checks, source identities, and any unavailable release gates. A direct Python notebook-cell replay is distinct from a native Jupyter kernel replay. Checksums establish byte identity, not theorem correctness or publication status.

8.3 Study route and remaining obligations

Lemma 4.3a is the closed-system site-budget refinement. Lemma 4.3b is its finite-dilation extension to a reduced subsystem, including the decoupled case in which the environment Hamiltonian drops out of the reduced trajectory. Neither lemma removes the global factor g. Lemma 4.3c is one jump trajectory with no finite autonomous Hermitian dilation. Its coefficient 6 is scalar. Lemma 4.3d gives the coefficient $2(1+k^2)G^2$ for a finite family of jumps and no Hamiltonian, where G bounds the sum of the squared operator norms of the jumps. On the Lemma 4.3c data this coefficient equals 10 and the half-speed window is $1/5$, against 6 and $1/3$. The first-derivative budget $2G$ is achieved on that example; the quadratic coefficient is not sharp there. Neither lemma replaces g by $\max_j g_j$. Jump supports do not enter Lemma 4.3d. Lemma 4.3d does not assert that a general jump trajectory lacks a finite autonomous Hermitian dilation. Lemma 4.3e places a Hamiltonian of operator norm at most g beside that jump family. The quadratic coefficient is $2(1+k^2)(g+G)^2$, from the triangle bound $\|\mathcal{L}(X)\|_1 \le 2(g+G)\|X\|_1$. The cross term inside $(g+G)^2$ is not removed. On the commuting witness with $g = G = 1$ and $k = 2$, the coefficient is 40 and the half-speed window is $1/20$, against the scalar coefficient 6 and the window $1/3$. The first-derivative budget $2(g+G)$ is not achieved there. Site budgets still do not enter, and the scalar coefficient 6 is not claimed for the family. Lemma 4.3f is the acceleration witness for that cross term. With $g = G = 1$,

$\|\rho''(0)\|_1 = 10$, strictly larger than $4(g^2+G^2) = 8$ and strictly smaller than $4(g+G)^2 = 16$. The orbit is a product, so $\nu = 0$ and $D(t) = 0$. Substituting g^2+G^2 into the acceleration estimate is false. Whether the defect coefficient $2(1+k^2)(g+G)^2$ can be replaced by $2(1+k^2)(g^2+G^2)$ through some other argument is not settled. Remaining obligations are a concrete Lean matrix model, fresh pinned Lean replay, native notebook-kernel replay, removal of the global factor from the quadratic coefficient, a mixed defect coefficient strictly smaller than $2(1+k^2)(g+G)^2$ by an argument other than the acceleration substitution ruled out in Lemma 4.3f, a site-budget form of the jump bound, a supplied sheaf coefficient system, and any proved observation map linking quantum residual dynamics to cochain invariants. These are stated as unfinished work.

9. References and related repository material

1. Lim, L.-H. *Hodge Laplacians on graphs.* arXiv:1507.05379 (2015; revised 2019). Primary manuscript. Background for the finite Hodge viewpoint; it does not establish a quantum-entanglement interpretation here.
2. Haegeman, J., Lubich, C., Oseledets, I., Vandereycken, B., and Verstraete, F. *Unifying time evolution and optimization with matrix product states.* Physical Review B 94, 165116 (2016). DOI; author preprint. Context for variational tangent-space evolution, not a proof of this product-density bound.
3. Hall, M. J. W. *Correlation Distance and Bounds for Mutual Information* (2013). Primary manuscript. Defines the same two-party correlation distance and gives the two-qubit sufficient entanglement criterion at distance greater than one.
4. Aaronson, B., Lo Franco, R., Compagno, G., and Adesso, G. *Hierarchy and dynamics of trace distance correlations.* New Journal of Physics 15, 093022 (2013). Primary manuscript. Distinguishes the closest product state from the product of marginals.
5. Carroll, J. H. *Projection Obstruction repository series.* Repository collection. Related working papers motivate residual/cochain separation and projection-memory diagnostics. Their versions and deposition identities are not inferred from this package.

Supplemental formalization: uncertainty and correction (23 October 2026)

The standalone supplemental build `python tools/replay_projection_bounds.py` compiles `lean4/ProjectionBounds.lean` with Lean 4.35.0 and replays every named theorem in `lean4/ProjectionCensus.lean`. The 20-theorem result and exact source hashes are recorded in `lean4/projection_axiom_census.json`; the text axiom output is `lean4/projection_axiom_report.txt`. This supplements the existing census without replacing its historical scope. Dependencies, source pins and licenses are shipped in `lean4/projection_vendor/`; no sibling paper or network fetch is needed. Only `propext`, `Quot.sound` and `Classical.choice` are allowed.

Bound	Lean result	Explicit premise retained
Robust rate and onset	`lowerRate_le`, `uncertain_onset`, `half_slope_window`	A two-sided rate enclosure and the actual Taylor lower estimate on its horizon
Correction deadline	`correction_window_iff`, `zero_rate_window`, `safe_horizon`, `exhausted_budget`	Positive rate for division; a separate zero-rate branch; independently valid budgets and horizon

Bound	Lean result	Explicit premise retained
Actual corrected update	`corrected_step`, `accumulated_budget`	A nonexpansive evolution, residual enclosure and correction displacement bound
Finite transported jumps	`transported_jump_bound`	The variation-of-constants identity, an integral bound, and contractive transports
Product and observed residual uncertainty	`factor_replacement_bound`, `product_residual_uncertainty`, `product_defect_uncertainty`, `observed_residual_bound`, `observed_defect_interval`	A factor-replacement path with each step bounded by the state distance; a declared Lipschitz observation

The scalar results are over the real numbers. The jump and residual theorems are over arbitrary normed additive groups, so they apply to a trace-norm carrier once its physical obligations are proved. In particular, the constant $k + 1$ is derived by telescoping k factor replacements and adding the original state discrepancy. The finite sum of jump norms is derived from the supplied transported-jump identity. Lean does not construct the continuous propagator, prove the density-matrix tensor/partial-trace estimates, or certify numerical integration in this supplement. Those statements remain written analytic premises. The new bounds do not turn a supplied error radius into an observed or statistically justified radius.

Three compiled examples check the exact allowed budget, a strictly excessive budget, and a rate completely explained by uncertainty. The companion Python examples test exact arithmetic on supplied budgets; the fresh Lean replay checks the general implications above.